

CONCEPT BRIEF

2D Space MORPG

Fog. Energy. Ships. Risk.

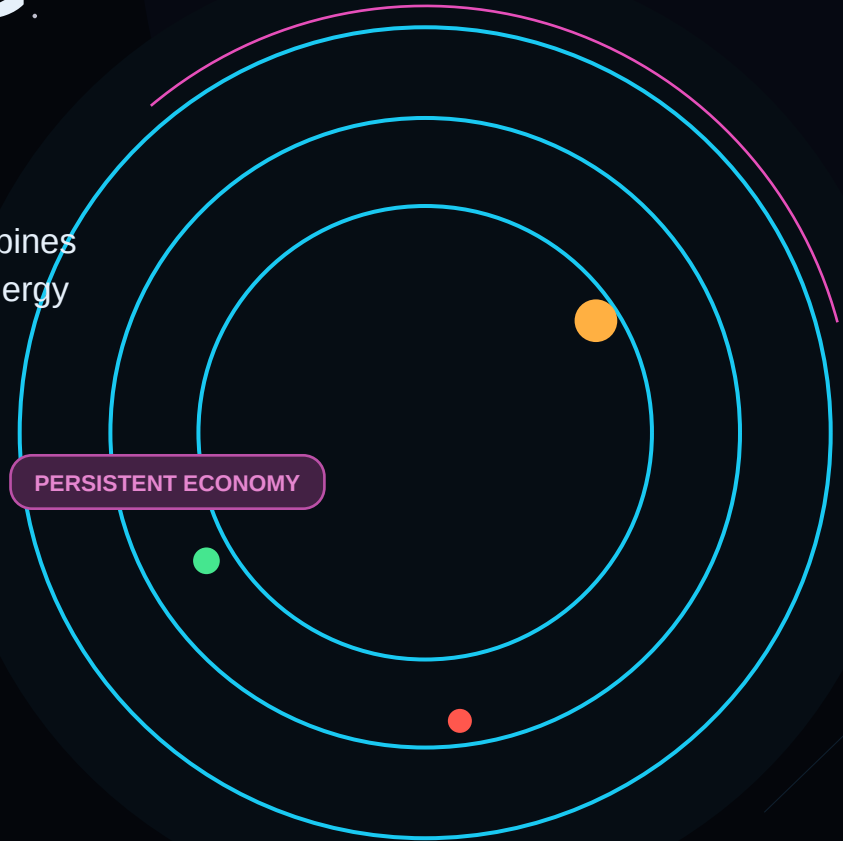
A server-authoritative browser and desktop space game that combines real-time ship control, radar-gated fog of war, planet discovery, energy economy, PvP risk, and DarkOrbit-like ship progression.

BROWSER + DESKTOP

SERVER AUTHORITATIVE

PIXEL SPACE CANVAS

PERSISTENT ECONOMY



The Pitch

A real-time space game where information is the most valuable resource.

Players fly one active ship through a fog-covered galaxy, discover planets and resources, build an energy network, upgrade their hangar, and risk cargo in dangerous PvP sectors.

Radar-Gated Interaction

You cannot attack, loot, gather, or even know about entities unless your ship or network can detect them.

Energy Economy

Discovered planets and stations produce energy that powers crafting, laser ammo, wormholes, upgrades, and strategic expansion.

Ship Identity

Every chassis has different slots, stats, passive bonuses, and saved loadouts. One active ship, many hangar choices.

Risk Zones

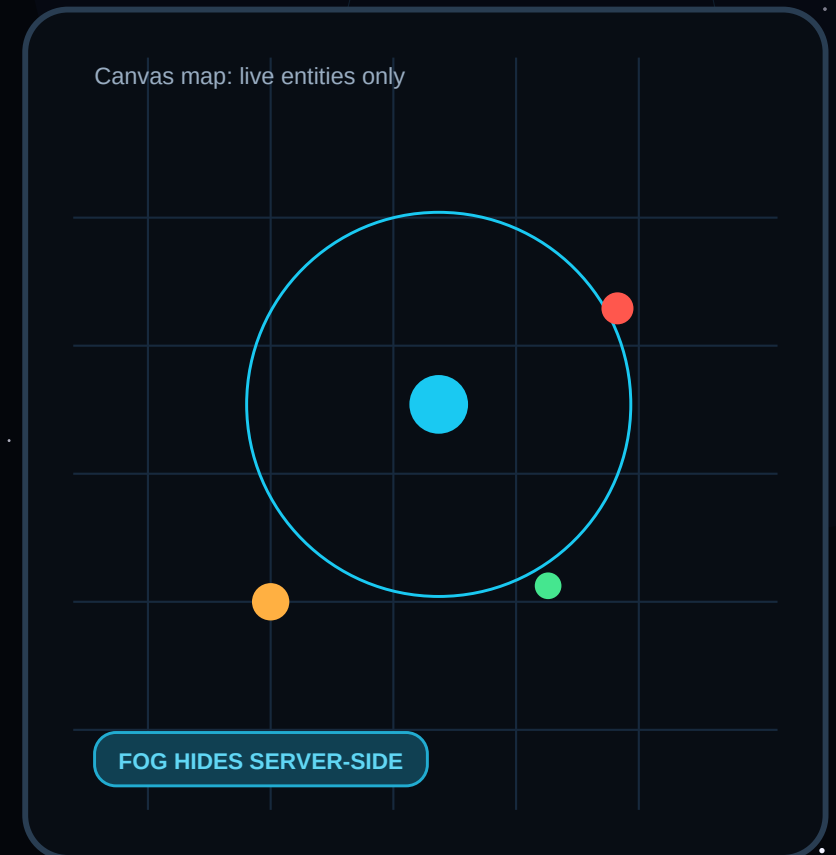
Safe space teaches the game. Deep space pays better, enables PvP, and can drop cargo when you die.

Core Game Loop



Moment-to-Moment Play

- Click or command the ship to move across a 2D space map.
- Scan fog, classify signals, and reveal planets, NPCs, loot, or enemy ships.
- Choose tasks such as mining, salvaging, scanning, hauling, or fighting.
- Use default auto-attack for farming, then manually trigger rockets, modules, scanners, or defensive tools during harder fights.
- Return to stations or planets to sell cargo, repair ships, swap loadouts, and craft upgrades.



Fog of War as a Rule

The server has the truth. Each client receives only the entities and intel that player is allowed to know. If radar cannot detect it, the browser never receives it.

Unknown

No data is sent. The client cannot inspect, target, loot, or infer the entity.

Weak Signal

Approximate contact. Maybe a planet, ship, loot, anomaly, or noise.

Visible

Live authoritative updates: position, velocity, HP, shield, state.

Last Known

Stale intel after contact is lost. Useful, but no longer live truth.

Visibility Pipeline

- Spatial grid candidates -> exact distance -> sensor strength -> stealth/signature -> nebula/jammer penalties -> line of sight -> per-player replication delta.
- Interactions such as attack, loot, gather, and portal use are rejected unless the relevant entity is currently visible or sensor-locked.

Planets and Energy

Exploration is not cosmetic. New planets unlock production, crafting paths, radar coverage, and long-distance logistics.

Planetary Energy

- Produced by claimed or developed planets.
- Spent on laser ammo production, buildings, upgrades, crafting, wormhole upkeep, station boosts, and strategic projects.
- Creates a reason to explore and defend territory.



Ship Capacitor

- Local combat energy pool on the active ship.
- Spent by lasers, shields, radar pulses, warp bursts, jammers, and active modules.
- Determined by chassis, generators, modules, and buffs.

Key balance idea: planets help you build and sustain power, but combat is still limited by the ship's capacitor and fitted modules. Economy fuels progression; moment-to-moment fights remain tactical.

Ships, Hangar, and Roles

The player owns multiple ships but deploys one active ship at a time. Each chassis has different slots, stats, passives, and saved loadouts.

Support Relay

Energy regen aura, radar support, fleet sustain.

Bulwark Hauler

High shield and cargo, few lasers, slow but valuable.

Interceptor

Fast, armed, low cargo. Built for PvP hunters.

Explorer

Strong radar, anomaly detection, fog scouting.

Industrial Miner

Mining bonuses, cargo modules, weak in combat.

Loadout Promise

- A more expensive ship is not automatically best at everything. It unlocks a different identity: more laser slots, stronger shields, bigger cargo, radar dominance, support auras, or specialist utility.
- If your active ship is destroyed, it must be repaired. You can deploy another hangar ship while waiting, which keeps death meaningful without ending the session.

Combat and PvP Risk

Attack Validation

- Target visible or sensor-locked.
- Weapon equipped and cooldown ready.
- Range, line of sight, safe-zone, and faction rules pass.
- Hit chance, shield, armor, and damage calculated server-side.

Auto + Manual Combat

- Default lasers can auto-fire for farming.
- Manual rockets, modules, scans, jammers, shields, and burst skills create skill expression.
- Energy/capacitor pressure becomes the combat rhythm.

Safe

Low reward, no open PvP.

Contested

PvP enabled, better resources.

Deep Space

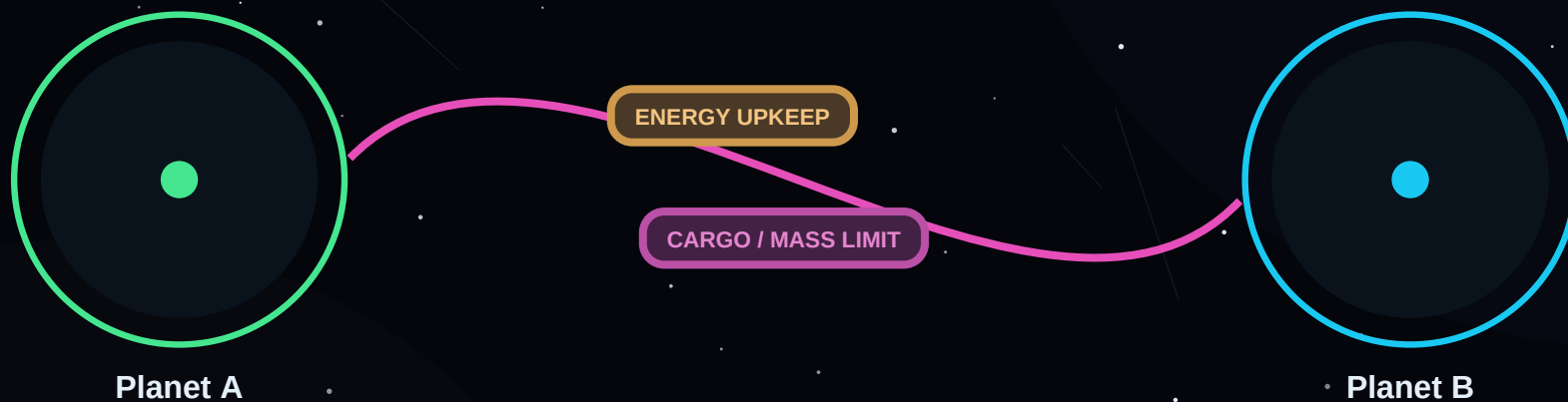
Higher reward, cargo drop risk.

Dead Space

Radar distortion, rare finds, highest danger.

Wormholes and Logistics

Discovered planets can become strategic anchors. Wormholes are not free teleporters; they are expensive logistics links that consume energy and create territory value.



Strategic Use

- Build routes between claimed or discovered worlds.
- Move faster across your own network, but pay upkeep and expose infrastructure.
- PvP groups can raid supply routes, disrupt anchors, or fight over high-value planets.

Server Architecture



Portal transitions are logical routing changes. The socket can stay connected while the player moves from Map A to Map B.

First Vertical Slice

The first prototype should prove the core architecture before adding content volume.

- One gateway, one server process, two maps, one portal.
- One active player ship with server-authoritative movement.
- One NPC enemy with range, damage, death, respawn, and loot drop.
- One resource node with gather ticks, cargo limit, and respawn.
- Server-side fog: hidden entities are never sent to the client.
- PixiJS canvas client with ship, radar radius, fog overlay, loot, and panels.

Success Criteria

- A player can explore unknown space.
- Detection controls what appears on screen.
- Portal travel moves the player between maps.
- Combat, loot, and gathering are all validated by the server.
- The prototype already feels like the final game loop in miniature.

Inspirations and Differentiation

Dark Forest

Cryptographic fog of war, exploration, planets, artifacts, wormholes. We borrow the information-war fantasy without using blockchain.

DarkOrbit

Readable 2D space combat, portals, ship identity, lasers, PvE/PvP farming loops. We make it more server-authoritative and exploration-heavy.

OGame / Travian

Long-term browser strategy, resource loops, alliances, timers, and macro progression. We add a live ship and tactical map layer.

EVE Online / EVE Echoes

Mining, hauling, PvP risk, player economy, space roles. We keep the client lighter and more approachable.

Reference URLs: darkforest.zkga.me/blog, darkorbit.com, gameforge.com/ogame, kingdoms.com, eveonline.com